

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:18-01-25**
Time: 3HRS **Max. Marks: 300**

Name of the Student: _____ **H.T. NO:** | | | | | | | |

18-01-25_SR.* CO-SUPER CHAINA (MODEL-A& APEX)_JEE-MAIN_GTM-20_SYLLABUS

PHYSICS: Thermodynamics, Mechanics and Applications, Optics, Electromagnetism, Modern Physics

CHEMISTRY: d & f block elements, Isomerism & Co-ordination chemistry, Chemical Kinetics, Redox titrations, Thermo Chemistry, Common ion effect on solubility, Practical Organic Chemistry, Work done in cyclic process, Nucleophilicity order, Biomolecules, Chemical bond and Periodic Properties, Questions based on General Characteristics of s, P-block, Qualitative Analysis (NCERT) - Dry Tests, Amines, Purification and Characterisation, Nomenclature (NCERT Applications), Applications of Coordination Components, Physical Chemistry- Graphs, Batteries

MATHEMATICS: Sets & Relations, Functions, Continuity & Differentiability, Indefinite Integration, Differential Equations, Binomial Theorem, Matrices & Determinants, Statics & Random variables, Definite Integration (Inequalities)

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A material has zero magnetic moment in all of its atoms. This material if placed in external magnetic field of magnitude B_{ext} the magnitude of magnetic field inside the material becomes B_{in}. Then mark the correct option

   A) $B_{in} > B_{ext}$ B) $B_{in} < B_{ext}$

   C) $B_{in} = B_{ext}$ D) Comparison depends on temperature

2. An ideal solenoid has some current, causing a magnetic intensity of 10^{-2} in SI units inside the solenoid. If a ferromagnetic material of relative permeability 150 is inserted in the solenoid. The magnetic intensity in solenoid after the insertion of the material is

   A) 10^{-2} SI units B) $\frac{10^{-2}}{150}$ SI units C) 150×10^{-2} SI units D) 149×10^{-2} SI units

3. A sample of gas is confined in a closed chamber. This gas undergoes following processes one after the other:

   Process I: It is heated to increase its temperature by 50K with its volume remaining constant.

   Process II: Its volume is adiabatically doubled.

   Mark the correct option

   A) Mean free path during the process I increases

   B) Mean free path during the process I decreases

   C) Mean free path during the process II increases

   D) Mean free path during the complete sequence of processes I and II remains constant.

4. Mass of a particle is 2.00 kg. Its velocity relative to the ground is 2.000 m/s. Velocity of an observer is 1.50 m/s in the same direction as that of the particle. The calculated value of the magnitude of momentum of the particle relative to the ground is reported as p_G, and that relative to the observer as p_O. Mark the correct option:

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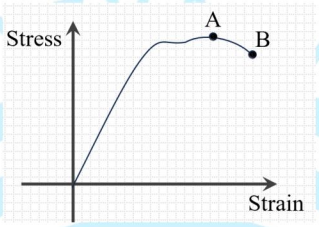
A) $p_G = 4.0$ kgm/s and $p_O = 1.0$ kgm/s B) $p_G = 4.00$ kgm/s and $p_O = 1.0$ kgm/s

C) $p_G = 4.0$ kgm/s and $p_O = 1.00$ kgm/s D) $p_G = 4.00$ kgm/s and $p_O = 1.00$ kgm/s

5. The dimensional formula of yield strength is

A) $[ML^{-1}T^{-2}]$ B) $[M^0L^1T^{-2}]$ C) $[M^{-1}L^1T^2]$ D) $[ML^{-1}T^2]$

6. Figure shows stress strain relation for a uniform wire. Points A and B are shown in graph.



From the graph, we can conclude that

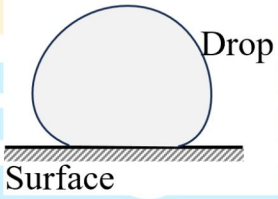
A) If gap between A and B is less, material is more brittle

B) If gap between A and B is less, material is more ductile

C) If point A is higher, ultimate strength of material is less

D) In the region AB, material is behaving as elastic material

7. Figure shows a liquid drop spread on a flat horizontal surface.



The angle of contact of this liquid with the surface is

A) Zero B) Acute angle C) 90° D) Obtuse angle

8. "During the day, the ground heats up more quickly than large bodies of water do. This occurs both because water has a greater specific heat capacity and because mixing currents disperse the absorbed heat throughout the great volume of water."

This phenomenon leads to

A) The flow of air from water body side to landmass side

B) The flow of air from landmass side to water body side

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C) The upward flow of air in water body side

D) The downward flow of air in landmass side

9. Which of the following statement is not a form of the second law of thermodynamics

A) No process is possible whose sole result is the absorption of heat from a reservoir and the complete conversion of the heat into work

B) No process is possible whose sole result is the transfer of heat from a colder object to a hotter object.

C) No engine can have efficiency greater than that of the Carnot engine.

D) The energy supplied to the system goes in partly to increase the internal energy of the system and the rest in work on the environment

10. Consider a gas of Carbon mono-oxide, the molecules of which can undergo vibrations along with translation and rotations. The molar heat capacity of this gas at constant volume is

A) R B) $2.5R$ C) $3R$ D) $3.5\ R$

11. The law of equipartition of energy states that:

"The energy for each degree of freedom in thermal equilibrium is $\frac{1}{2}k_BT$."

Here:

A) Each linear term or quadratic term (in terms of motion variables) in the total energy expression of a molecule is to be counted as a degree of freedom.

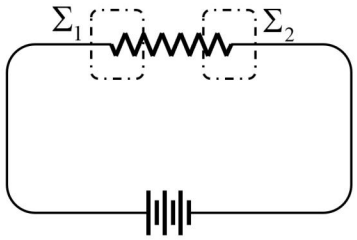
B) Each potential energy term in the total energy expression of a molecule is to be counted as a degree of freedom.

C) Each quadratic term (in terms of motion variables) in the total energy expression of a molecule is to be counted as a degree of freedom.

D) Each vibrational mode in the total energy expression of a molecule is to be counted as a degree of freedom.

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12. Figure shows an electrical circuit with a non-zero resistance R connected to a battery of non-zero *emf* E. The connecting wires are resistance less.



When the current is present in the circuit, two cubical Gaussian surfaces Σ_1 and Σ_2 are considered as shown by dotted lines. Then

A) Σ_1 and Σ_2 contain positive charges

B) Σ_1 and Σ_2 contain positive and negative charges respectively

C) Σ_1 and Σ_2 contain negative and positive charges respectively

D) Neither Σ_1 nor Σ_2 contain any net charge

13. Which of the following phenomena is not related to surface tension of water:

A) It is difficult to lift up a coin placed on wet surface in comparison to dry surface

B) Two wet hairs have tendency to stick together, although dry hairs do not

C) Liquids have a well-defined free surface while gases do not

D) If a coin and a drop of water starts falling from height of clouds, drops of water fail to achieve velocity equal to coin as they land on ground

14. If v_s, v_x and v_m are the speeds of gamma rays, X-rays and microwaves respectively in vacuum, then

A) $v_s > v_x > v_m$ B) $v_s < v_x < v_m$ C) $v_s < v_x < v_m$ D) $v_s = v_x = v_m$

15. A light beam is incident parallel to principal axis of a converging lens at a distance h from the principal axis. The value of h is assumed much smaller compared to focal length of the lens. The light undergoes a total deviation of δ in passing through the lens. The converging power of the lens is

A) $\frac{h}{\tan\delta}$ B) $\frac{1}{\delta}$ C) $\frac{\tan\delta}{h}$ D) $\frac{1}{h\delta}$

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16. Consider a telescope whose objective has a focal length of 100 cm and the eyepiece a focal length of 1 cm, used under normal adjustment. A pair of stars having actual separation 1' (one minute of arc which is equivalent to $\frac{1}{60}\times 1^{\circ}$), is viewed through this telescope. The stars will appear at an angular separation

A) 20' B) 1° C) 1.33° D) 1.67°

17. Mark the correct option regarding the reflecting telescope:

A) The incident light coming from a large distance, first falls on a lens and then on a mirror

B) It shows chromatic aberration

C) It doesn't show chromatic aberration

D) It has very less achievable magnification in comparison to refracting telescopes

18. Which of the following phenomena doesn't involve total internal reflection of light:

A) Transmission of electromagnetic signal through an optical fiber

B) Brilliance shown by diamond

C) Formation of rainbow in sky

D) Sky appearing blue

19. Polarisation of light establishes

A) corpuscular theory of light B) quantum nature of light

C) transverse nature of light D) all the three

20. Which of the following rays get emitted by a human body?

A) X-rays B) UV rays C) Visible rays D) IR rays

SAMAYEVA JAYATHE
THE NARAYANA GROUP

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
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21. The magnetic susceptibility of a paramagnetic material at $-73°C$ is 0.0075 and at another temperature $-173°C$ is $(x)\times10^{-4}$. Find the value of x

22. In a Young's double-slit experiment, the intensity ratio of maxima and minima is infinite. The ratio of the amplitudes of two sources is ____.

23. The magnetic field of an Electromagnetic Wave is given by
$B_Y = \left(3\times10^{-7}\,\text{T}\right)\sin\left(\left(\pi\times10^{3}\,\text{m}^{-1}\right)x+\left(6.29\times10^{12}\,\text{Hz}\right)t\right)$.
The wavelength of the Electromagnetic Wave in mm is _____.

24. We can use the law of equipartition of energy to determine specific heats of solids. Consider a solid of large number of identical atoms, each vibrating about its mean position. If the vibrations occur in 3 dimensions, the molar heat capacity of the solid is xR. The closest integer that represents x is _____.
[Ignore the exception of Carbon in solid form]

25. Two trains one of 100 m and another of length 125m, are moving in mutually opposite directions along parallel lines, meet each other, each with speed 10m/s. If their acceleration are 0.3m/s^2 and 0.2m/s^2 respectively, then the time taken to pass each other in seconds will be ____.

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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26. S_1: Mn^{2+} compounds are more stable than Fe^{2+} towards oxidation to their +3 state.

S_2: Titanium and copper both in the first series of transition metals exhibits +1 oxidation state frequently.

S_3: Cu^{+} ion is stable in aqueous solutions.

S_4: The E^{θ} value for the Mn^{3+}/Mn^{2+} couple is more positive than that for Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+}

A) T T F T B) T F F T C) T F T T D) F F T F

27. VO_2 is an amphoteric oxide and in acidic medium it forms

A) VO^{2+} B) VO_2^{+} C) V^{3+} D) VO_2^{2+}

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28. Arrange the following in order of decreasing number of unpaired electrons

I) $\left[Fe(H_2O)_6\right]^{2+}$ II) $\left[Fe(CN)_6\right]^{3-}$ III) $\left[Fe(CN)_6\right]^{4-}$ IV) $\left[Fe(H_2O)_6\right]^{3+}$

A) IV, I, II, III B) I, II, III, IV C) III, II, I, IV D) II, III, I, IV

29. What volume of 0.40 M-$Na_2S_2O_3$ would be required to react with the I_2 liberated by adding excess of KI to 50 ml of 0.20 M $CuSO_4$?

A) 12.5 ml B) 25 ml C) 50 ml D) 2.5 ml

30. For a first order reaction $A \rightarrow B + 2C + 3D$ (A is optically inactive and B, C and D are dextrorotary), the optical rotation at time t and ∞ ae r_t and r_∞ respectively, the expression for rate constant is

A) $K = \frac{1}{t}\ln\frac{r_t}{r_\infty - r_t}$ B) $k = \frac{1}{t}\ln\frac{r_\infty}{r_\infty - r_t}$

C) $k = \frac{1}{t}\ln\frac{r_\infty - r_t}{r_t}$ D) None of these

31. The rate equation for the reaction $2A + B \longrightarrow C$ is found to be: rate = k [A] [B]. The correct statement in relation to this reaction is that the

A) unit of k must be sec^{-1}

B) $t_{1/2}$ is a constant

C) rate of formation of C is twice the rate of disappearance of A

D) value of k is independent of initial concentrations of A and B.

32. ΔH^o_f of water is –285.5 $kJ\,mol^{-1}$ if enthalpy of neutralisation of monoacidic strong base with strong acid is –57.3 $kJ\,mol^{-1}$ ΔH_f^o of OH^- ion will be

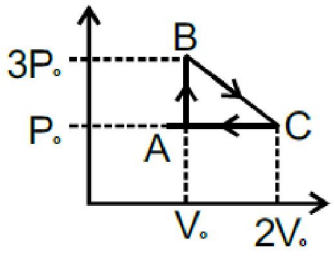
A) $-228.5 kJ\,mol^{-1}$ B) $228.5 kJ\,mol^{-1}$

C) $114.5 kJ\,mol^{-1}$ D) $-114.5 kJ\,mol^{-1}$

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33. One mole of ideal monoatomic gas is carried through the reversible cyclic process as shown in figure. Calculate net heat absorbed by the gas in the path BC.



A) $\frac{1}{2}P^oV^o$ B) $\frac{7}{2}P^oV^o$ C) $2P^oV^o$ D) $\frac{5}{2}P^oV^o$

34. The lattice enthalpy and enthalpy of solution in water for solid NaCl are 753 kJ mol^{-1} and 5 kJ mol^{-1} respectively. If the solution enthalpies of Na^+ and Cl^- are in the ratio 6 : 5, the enthalpy of hydration of Na^+ ion is.

A) $408\,kJ\,mol^{-1}$ B) $-412\,kJ\,mol^{-1}$ C) $-408\,kJ\,mol^{-1}$ D) $-412\,kJ\,mol^{-1}$

35. $SrCO_3\left(K_{sp}=10^{-10}\right)$ and $ZnCO_3\left(K_{sp}=1.5\times10^{-11}\right)$ are dissolved together in a solution. The ratio of $\left[Sr^{2+}\right]/\left[Zn^{2+}\right]$ in the solution is

A) $\frac{10}{3}$ B) $\frac{3}{10}$ C) $\frac{20}{3}$ D) $\frac{3}{20}$

36. Solubility of $Fe(OH)_3\left(K_{sp}=10^{-38}\right)$

A) In a buffer solution of pH = 2 is 10^{-2} mol/L

B) In a buffer solution of pH = 11 is 10^{-29} mol/L

C) Both (A) & (B)

D) None of these

37. Arrange solubility of AgBr in the given solutions in increasing order : (consider no hydrolysis of Ag^+ ion)

(i) 0.1 M NH_3 (ii) 0.1 M $AgNO_3$ (iii) 0.2 M NaBr (iv) pure water

A) (iii) < (ii) < (iv) < (i) B) (iii) < (ii) < (i) < (iv)

C) (iii) < (ii) = (i) < (iv) D) (ii) < (iii) < (iv) < (i)

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38. In an alkaline energy cell the overall cell reaction is as follows:

$Zn(s)+2MnO_2(s)+2H_2O \rightarrow Zn(OH)_2(s)+2MnO(OH)$

Which of the following reactions is taking place at the cathode?

A) $2MnO_{2(s)}+2H_2O+2e \rightarrow Zn(OH)_{2(s)}+2MnO(OH)_{(s)}$

B) $2MnO_{2(s)}+2H_2O+2e \rightarrow 2MnO(OH)_{(s)}+2OH^-_{(aq)}$

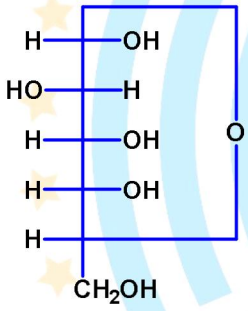
C) $Zn_{(s)}+2OH^-_{(aq)} \rightarrow Zn(OH)_{2(s)}+2e$

D) $Zn(OH)_{2(s)}+2e \rightarrow Zn_{(s)}+2OH^-_{(aq).}$

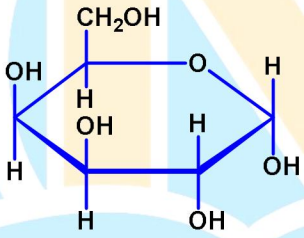
39. x g of KHC_2O_4 requires 100 ml of 0.02 M-$KMnO_4$ in acidic medium. In another experiment, y g of KHC_2O_4 requires 100 ml of 0.05 M-$Ca(OH)_2$. The ratio of x and y is

A) 1 : 1 B) 1 : 2 C) 2 : 1 D) 5 : 4

40. Which among the following can give saccharic acid on oxidation with HNO_3?



A)



B) C) Gluconic acid D) All

41. Identify the correct one letter code of naturally occurring aromatic ring containing amino acids from the from the given below

A) FYPH B) WHFI C) YFHW D) HWTY

42. If x is the number of isomeric compounds with molecular formula C9H10O which (i) do not dissolve in NaOH (ii) do not dissolve in HCl (iii) do not give orange PPT with 2,4-DNP (iv) on hydrogenation give identical compound with molecular formula $C_9H_{12}O$. What is the value of x?

A) 4 B)2 C) 5 D) 6

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43. Observe the below given order

i) $NH_3 > CH_3COO^-$ ii) $NO_2^- > OH^-$ iii) $Cl^- > SH^-$ iv) $CN^- > OH^-$

Select the correct one

A) i) B) i) & iv) C) i), iii) & iv) D) i), ii), iii) & iv)

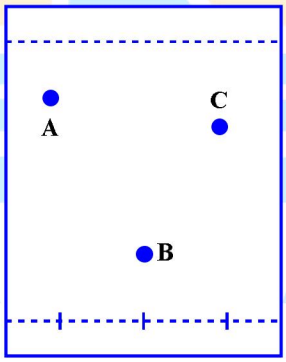
44. The correct match between item 'I' and item 'II' is:

	Item-I		**Item-II**
(A)	Adenine	(P)	$CHCl_3$ + aq KOH
(B)	Serine	(Q)	Ceric ammonium nitrate
(C)	Keratin	(R)	Oxidation with Br2 water
(D)	2-Deoxyribose	(S)	Biuret test

(A) A) → P; B) → R; C)→S; D) → P (B) A) → S; B) → Q; C)→ P; D) → S

(C) A) → R; B) → P; C)→ S; D) →P (D) A) → P; B) → Q; C)→ S; D) → R

45. Thin layer chromatography of a mixture shows the following observation



Select the **CORRECT** statement about compound A, B & C

A) Rf (retardation factor) value of A & C are identical

B) During the elution spot B will come first from silica gel column

C) During the elution spot A will come first from silica gel column

D) All

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CTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

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46. How many of given statements are **TRUE** for lanthanides?

(1) The common oxidation state of all lanthanides is +3.

(2) All lanthanides belongs to group III of modern periodic table.

(3) All lanthanides belongs to seventh period of periodic table.

(4) The radius for all lanthanides cations (M^{+3}) decrease from Ce^{+3} to Lu^{+3}.

(5) The basic character of hydroxide increases on moving from $Ce(OH)_3$ to $Lu(OH)_3$.

(6) All lanthanides are metals.

(7) The reducing power of lanthanides decreases from Ce to Lu.

(8) Lanthanides reacts with cold water to liberate H_2.

47. Calculate total number of geometrical, optical and structural isomers of the compound $\left[Rh(en)_2(NO_2)_2\right]NO_3$

48. The $\Delta H^0{}_f\left(K_{(F)},s\right)$ is –563 $kJ\,mol^{-1}$. The ionization enthalpy of K(g) is 419 $kJ\,mol^{-1}$. And the enthalpy of sublimation of potassium is 88 $kj\,mol^{-1}$. The electron affinity of F(g) is 322 $kJ\,mol^{-1}$ and F–F bond enthalpy is 158 $kJ\,mol^{-1}$. Calculate the magnitude of lattice enthalpy of KF(s).

The given data are as follows:

i) $K(s)+\frac{1}{2}F_2(g)+KF(s)$ $\quad \Delta H^0{}_f=-563\,kJ\,mol^{-1}$

ii) $K(g)\rightarrow K^+(g)+e^-$ $\quad \Delta H^0{}_{Ioniz}=419\,kJ\,mol^{-1}$

iii) $K(s)\rightarrow K(g)$ $\quad \Delta H^0{}_{sub}=88\,kJ\,mol^{-1}$

iv) $F(g)+e^-\rightarrow F^-(g)$ $\quad \Delta H^0{}_{eg}=-322\,kJ\,mol^{-1}$

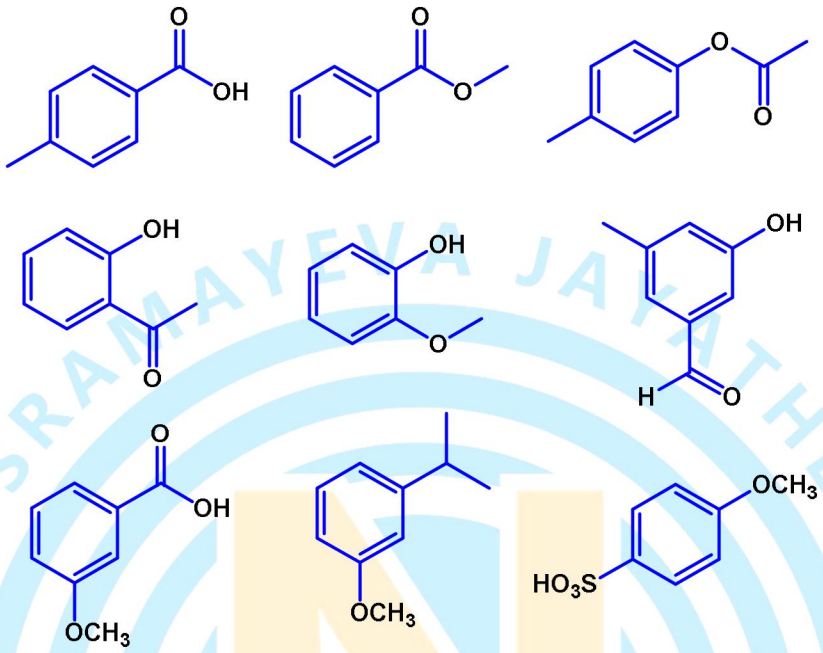
v) $F_2(g)\rightarrow 2F(g)$ $\quad \Delta H^0{}_{diss}=158\,kJ\,mol^{-1}$

vi) $K^+(g)+F^-(g)\rightarrow KF(s)$ $\quad \Delta H^0{}_L=?$

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49. If (p) is the number of compounds which are soluble in cold $NaHCO_3$; (q) is the number of compounds which are soluble in cold NaOH and (r) is the number of compounds which are soluble in hot NaOH solution then what is the value of p + q + r ?



50. How many of the following statements are **CORRECT** regarding Chromatography

i) In chromatography technique the purification of compound is dependent on physical state of pure compound

ii) In chromatography R_f (retardation factor) value is independent on the mobile phase.

iii) Chromatography technique can be used to test the purity of the compounds.

iv) In chromatography technique stationary phase can be either liquid or gas

v) Thin layer chromatography (TLC) can be used for the separation of substances

vi) In chromatography technique gas can be used as mobile phase

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

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51. Let $A=\{n\in N: n \text{ is a } 3-\text{digit number}]$ $B=\{9k+2: k\in N\}$ and $C=\{9k+\ell: k\in N\}$ for some $\ell(0<\ell<9)$ If the sum of all the elements of the set $A\cap(B\cup C)$ is 274×400, then ℓ is equal to

A) 1 B) 3 C) 5 D) 7

52. Let R_1 and R_2 be relations on the set $\{1,2,.....,50\}$ such that

$R_1=\{(p,p^n): p \text{ is a prime and } n\geq 0 \text{ is an integer}\}$ and

$R_2=\{(p,p^n): p \text{ is a prime and } n=0 \text{ or } 1\}$. Then, the number of elements in R_1-R_2 is

A) 2 B) 4 C) 6 D) 8

53. The set $S=\{1,2,3,.....,12\}$ is to be partitioned into three sets A, B, C of equal size. Thus $A\cup B\cup C=S$, $A\cap B=B\cap C=A\cap C=\phi$. The number of ways to partition S is

A) $\frac{12!}{(4!)^3}$ B) $\frac{12!}{(4!)^4}$ C) $\frac{12!}{3!(4!)^3}$ D) $\frac{12!}{3!(4!)^4}$

54. Let α be the constant term in the binomial expansion of $\left(\sqrt{x}-\frac{6}{x^{3/2}}\right)^n$, $n\leq 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to

A) 9 B) 36 C) 18 D) 72

55. If the sum of the series $\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2.3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2.3}+\frac{1}{2.3^2}-\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{2^3.3}+\frac{1}{2^2.3^2}-\frac{1}{2.3^3}+\frac{1}{3^4}\right)+....\text{is}\frac{\alpha}{\beta}$, where α and β are co-prime, then $\alpha+3\beta$ is equal to

A) 7 B) 14 C) 21 D) 28

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56. Let μ be the mean and σ be the standard deviation of the distribution

x_i	0	1	2	3	4	5
f_i	k+2	2k	k^2-1	k^2-1	k^2+1	$k-3$

where $\sum f_i = 62$. If $[x]$ denotes the greatest integer $\le x$, then $[\mu^2+\sigma^2]$ is equal to

A) 8 B) 7 C) 6 D) 9

57. Let $A=\{1,3,7,9,11\}$ and $B=\{2,4,5,7,8,10,12\}$. Then the total number of one-one maps $f: A \to B$, such that $f(1)+f(3)=14$, is:

A) 180 B) 120 C) 480 D) 240

58. Let $A=\{2,3,4\}$ and $B=\{8,9,12\}$. Then the number of elements in the relation $R=\{((a_1,b_1),(a_2,b_2)) \in (A\times B, A\times B) : a_1 \text{ divides } b_2 \text{ and } a_2 \text{ divides } b_1\}$ is :

A) 36 B) 12 C) 18 D) 24

59. The number of functions f from $\{1,2,3,....,20\}$ onto $\{1,2,3,....,20\}$ such that $f(k)$ is a multiple of 3, whenever k is a multiple of 4 is :

A) $6^5 \times (15)!$ B) $5! \times 6!$ C) $(15)! \times 6!$ D) $5^6 \times 15$

60. Let $M=\left\{A=\begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \{\pm 3, \pm 2, \pm 1, 0\}\right\}$. Define $f: M \to Z$, as $f(A) = \det(A)$, for all $A \in M$, where Z is set of all integers. Then the number of $A \in M$ such that $f(A)=15$ is equal to

A) 4 B) 8 C) 16 D) 32

61. Let the determinant of a square matrix A of order m be m – n, where m and n satisfy $4m+n=22$ and $17m+4n=93$. If $\det\left(n\,\text{adj}\left(\text{adj}\left(mA\right)\right)\right) = 3^a 5^b 6^c$, then $a+b+c$ is equal to:

(Here det. represents determinant of a square matrix)

A) 96 B) 101 C) 109 D) 84

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62. Let $f(x)=[x^2-x]+|-x+[x]|$, where $x \in R$ and [t] denotes the greatest integer less than or equal to t. Then, f is

A) continuous at x = 0, but not continuous at x = 1

B) continuous at x = 0 and x = 1

C) not continuous at x = 0 and x = 1

D) continuous at x = 1, but not continuous at x = 0

63. The number of points where the function

$$f(x)=\begin{cases}(2x^2-3x-7) & \text{if} \quad x \le -1 \\ [4x^2-1] & \text{if} \quad -1<x<1 \\ |x+1|+|x-2| & \text{if} \quad x \ge 1\end{cases}$$

[t] denotes the greatest integer $\le t$, is discontinuous is _____.

A) 5 B) 14 C) 21 D) 7

64. If $I=\int_1^2 \frac{dx}{\sqrt{2x^3-9x^2+12x+4}}$, then:

A) $\frac{1}{8}<I^2<\frac{1}{4}$ B) $\frac{1}{9}<I^2<\frac{1}{8}$

C) $\frac{1}{16}<I^2<\frac{1}{9}$ D) $\frac{1}{6}<I^2<\frac{1}{2}$

65. Let the slope of the line $45x+5y+3=0$ be $27r_1+\frac{9r_2}{2}$ for some $r_1, r_2 \in \mathbb{R}$. Then

$\lim_{x\to 3}\left(\int_3^x \frac{8t^2\,dt}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}\right)$ is equal to _____

A) 6 B) 12 C) 24 D) 4

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66. If the solution $y = y(x)$ of the differential equation $(x^4 + 2x^3 + 3x^2 + 2x + 2)$

$dy - (2x^2 + 2x + 3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then y(0) is equal to:

A) $-\frac{\pi}{12}$ B) 0 C) $\frac{\pi}{4}$ D) $\frac{\pi}{2}$

67. Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}, 0 < x < \frac{\pi}{2} \text{ with}$$

$y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}$. If $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to

A) 1 B) 2 C) 3 D) 4

68. Let S_n be the sum to n – terms of an arithmetic progression 3, 7, 11, ……. .If

$40 < \left(\frac{6}{n(n+1)}\sum_{k=1}^{n} S_k\right) < 42$, then n equals ____

A) 9 B) 3 C) 27 D) 18

69. Let the mean and the variance of 20 observations $x_1, x_2, ...x_{20}$ be 15 and 9, respectively. For $\alpha \in R$, if the mean of $(x_1 + \alpha)^2, (x_2 + \alpha)^2, .., (x_{20} + \alpha)^2$ is 178, then the square of the maximum value of α is equal to _____

A) 2 B) 6 C) 8 D) 4

70. Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f : A \to B \mid 2 \in f(A) \text{ and } f \text{ is not one} - \text{one}\}$ is

A) 38 B) 19 C) 57 D) 81

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SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $A=\{n\in N\,|\,n^2\le n+10,000\}, B=\{3k+1\,|\,k\in N\}$ and $C=\{2k\,|\,k\in N\}$, then the sum of all the elements of the set $A\cap(B-C)$ is equal to

72. Let $x=\sin\left(2\tan^{-1}\alpha\right)$ and $y=\sin\left(\frac{1}{2}\text{tna}^{-1}\frac{4}{3}\right)$. If $S=\{\alpha\in R: y^2=1-x\}$, then $\sum_{\alpha\in S}16\alpha^3$ is equal to _____.

73. If $X=\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}, Y=\alpha I+\beta X+\gamma X^2$ and $Z=\alpha^2 I-\alpha\beta X+(\beta^2-\alpha\gamma)X^2, \alpha,\beta,\gamma\in\mathbb{R}$.

If $Y^{-1}=\begin{bmatrix}\frac{1}{5}&\frac{-2}{5}&\frac{1}{5}\\0&\frac{1}{5}&\frac{-2}{5}\\0&0&\frac{1}{5}\end{bmatrix}$, then $(\alpha-\beta+\gamma)^2$ is equal to _____.

74. Let $f(x)=\min\{[x-1],[x-2],....,[x-10]\}$ where [t] denotes the greatest integer $\le t$.

Then $\int_0^{10}f(x)dx+\int_0^{10}(f(x))^2dx+\int_0^{10}|f(x)|dx$ is equal to

75. The maximum value of the term independent of t in the expansion of

$\left(t^2x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}, x\ge 0$, is K, then 8K is equal to _____

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